Lecture 9
Solutions

Homework 9: Issues, CI, and a release

Solutions to Homework 9

Part A.

```
$ bash codeHints/L09/setup_hw9.sh
$ cd practice
$ git clone hw9_server.git hw9
$ cd hw9
$ bash test.sh
FAIL: got 'Helo, world'
$ sed -i.bak 's/Helo/Hello/' greet.sh && rm greet.sh.bak    # or edit by hand
$ bash test.sh
PASS
$ git commit -am "Fix greeting typo" -m "Closes #1"
$ mkdir -p .github/workflows .github/ISSUE_TEMPLATE
```

Create `.github/workflows/ci.yml` (spaces only):

```
name: CI
on:
  push:
    branches: [main]
  pull_request:
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Run the tests
        run: bash test.sh
```

Create `.github/ISSUE_TEMPLATE/bug_report.md` and `.github/pull_request_template.md`:

```
---
name: Bug report
about: Something does not work
---

## What happened?

## What did you expect?
```

```
## What does this change?

## Checklist
- [ ] The tests pass
- [ ] I read my own diff
```

```
$ git add .github/workflows
$ git commit -m "Add CI workflow"
```

```
$ git add .github/ISSUE_TEMPLATE .github/pull_request_template.md
$ git commit -m "Add templates"
$ git tag v0.1.0 $(git rev-list --max-parents=0 HEAD)
$ git tag -a v1.0.0 -m "First release"
$ git push origin main v0.1.0 v1.0.0
$ printf 'next-bugfix=1.0.1\nissue-closes-when=merged\n' > ../hw9_answers.txt
```

(Part A checks only the workflow's words; GitHub reports wrong indentation in Part B.)

9.3–9.4. The checker runs your `test.sh` on a copy of the server's `main`, as the workflow does on GitHub. The second `-m` is a separate paragraph of the message; GitHub finds the keyword anywhere in it.

9.5. Without `uses: actions/checkout` the machine has no files, so `bash test.sh` fails. The folder is `.github/workflows` (with an *s*).

9.7. The annotated tag has its own object with your message (`git show v1.0.0` shows a *Tagger*). Tags are pushed only when you name them or use `--tags`.

9.8. Fixing a bug in `1.0.0` gives the patch version `1.0.1`. `Closes #12` closes the issue when the change is *merged* into the default branch, not when the pull request is opened.

Part B.

```
(GitHub) New repository: ci-practice, Public, no README
$ git remote add github git@github.com:YOURNAME/ci-practice.git      # the SSH address
$ git push github main v0.1.0 v1.0.0
(GitHub) Actions tab: the "CI" workflow ran: green tick
$ printf 'repo=YOURNAME/ci-practice\n' >> ../hw9_answers.txt
(GitHub) Issues > New issue "Add a note to the README"  -> number 1
$ git switch -c readme-note
$ echo "Notes coming soon." >> README.md
$ git commit -am "Add a note to the README"
$ git push github readme-note
(GitHub) Open the pull request, description "Closes #1"; wait for the green check;
    Merge
(GitHub) Issue #1 is now closed
$ printf 'issue=1\n' >> ../hw9_answers.txt
(GitHub) Releases > Draft a new release > tag v1.0.0 > Generate release notes > Publish
(GitHub) Actions: wait until the new run on main (after the merge) is green, then run
    the checker
```

Replace `YOURNAME` with your GitHub user name. The checker needs the answers file to contain `repo=`, `issue=`, and a remote of `practice/hw9` that points at that repository. It then asks GitHub's public interface whether the issue is closed, whether a release exists for `v1.0.0`, and whether the latest Actions run on `main` succeeded. If the last check is red, open the run's log in the Actions tab.

Common mistakes.

- A *tab* in the workflow, or a folder `.github/workflow` instead of `workflows`.
- Creating Issue #1 *before* the first push: the commit message `Closes #1` then closes it at once, and the pull request closes nothing. Push first, then create the issue.
- Forgetting to push the tags: `git push origin main` sends no tags.
- Creating the repository with a README: the first push is rejected. Create it empty.
- A private repository: the checker cannot see it and reports "cannot find".
- Running the checker while the workflow run on `main` is still in progress: the check "latest run

is green" fails until it finishes. Wait and run it again.